

21/04/2025



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015

CODE-A



AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-8

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Chemistry, Physics and Biology having 45 questions each in Chemistry and Physics and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



CHEMISTRY

1. Match the catalyst in List-I with process given in List-II

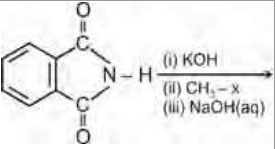
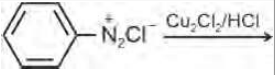
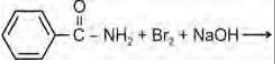
	Catalyst		Process
a.	V ₂ O ₅	(i)	Oxidation of SO ₂ in the manufacture of H ₂ SO ₄ by contact process
b.	PdCl ₂	(ii)	Polymerisation of alkynes
c.	Fe	(iii)	Wacker's process
d.	Nickel complexes	(iv)	Haber's process

Choose the correct answer

- (1) a(i), b(ii), c(iii), d(iv)
 (2) a(ii), b(i), c(iv), d(iii)
 (3) a(iii), b(iv), c(i), d(ii)
 (4) a(i), b(iii), c(iv), d(ii)
2. Choose the incorrect statement(s) for interstitial compounds.
 A. They have lower melting points than those of pure metals.
 B. They are very hard.
 C. They are chemically more reactive than pure metal
 D. They retain metallic conductivity
 Choose the correct option
 (1) A and C only
 (2) A and B only
 (3) A, C and D only
 (4) B only
3. Given below are two statements
Statement I: Magnitude of work done during isothermal expansion of an ideal gas is lesser than work done during adiabatic expansion of an ideal gas.
Statement II: Initial temperature is higher than final temperature during adiabatic expansion of an ideal gas.
 In the light of above statements, choose the correct answer from the options given below.
 (1) Both statements I and II are incorrect
 (2) Both statements I and II are correct
 (3) Statement I is incorrect but statement II is correct
 (4) Statement I is correct but statement II is incorrect
4. Mole fraction of solute in 1 molal aqueous solution will be
 (1) 0.17
 (2) 0.017
 (3) 0.83
 (4) 0.983

5. The molar heat capacity of water at constant pressure P is 75 JK⁻¹ mol⁻¹. When 1 kJ of heat is supplied to 100 g of water, the increase in temperature of water is
 (1) 0.24 K
 (2) 0.13 K
 (3) 1.2 K
 (4) 2.4 K
6. Crystal field stabilization energy for high spin d⁶ octahedral complex is (ignoring pairing energy)
 (1) -2.4 Δ₀
 (2) -1.6 Δ₀
 (3) -1.2 Δ₀
 (4) -0.4 Δ₀
7. The oxidation number of cobalt in K[Co(CO)₄] is
 (1) -1
 (2) +1
 (3) +3
 (4) -3
8. The correct decreasing order of basic strength of following amines in aqueous solution is
 (CH₃)₂NH (a) NH₃ (b) CH₃NH₂ (c) (CH₃)₃N (d)
 (1) (d) > (b) > (c) > (a)
 (2) (c) > (a) > (d) > (b)
 (3) (a) > (c) > (d) > (b)
 (4) (d) > (a) > (c) > (b)

9. Match List-I with List-II

	List-I (Name)		List-II (Chemical Reaction)
a.	Gabriel phthalimide synthesis	(i)	$\text{CH}_3 - \text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\Delta}$
b.	Sandmeyer reaction	(ii)	
c.	Hoffmann bromamide degradation reaction	(iii)	
d.	Carbylamine reaction	(iv)	

The correct match is

- (1) a(iii), b(iv), c(i), d(ii)
- (2) a(i), b(iii), c(ii), d(iv)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(iv), b(iii), c(ii), d(i)

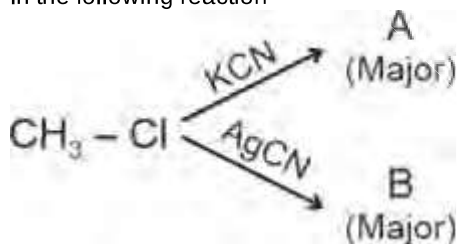
10. Given below are two statements

Statement I: 2-Bromopentane heated with alcoholic solution of KOH to form pent-2-ene as a major product

Statement II: Heating of 2-Bromopentane with alcoholic solution of KOH is an example of β -elimination reaction. In the light of above statements, choose the correct answer from options given below

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

11. In the following reaction



Products (A) and (B) respectively are

- (1) CH_3CN and CH_3NC
- (2) CH_3NC and CH_3CN
- (3) CH_3CN and CH_3CN
- (4) CH_3NC and CH_3NC

 12. The first ionisation enthalpy of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ is

- (1) +1.25 eV
- (2) -5.1 eV
- (3) -2.55 eV
- (4) +10.2 eV

13. Given below are two statements

Statement I: The solution which resist change in pH on dilution or with addition of small amount of acid or base is called buffer solution.

Statement II: A mixture of ammonium chloride and ammonium hydroxide acts as buffer solution. In the light of above statements, choose the correct answer from the options given below

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

14. In mercury cell, electrolyte used is

- (1) Paste of HgO and Carbon
- (2) Conc. NaOH
- (3) Conc. H_2SO_4
- (4) Paste of KOH and ZnO

15. Electronic configuration of an element having atomic number 60 (Neodymium) is

- (1) $[\text{Xe}] 4f^3 5d^2 6s^1$
- (2) $[\text{Xe}] 4f^3 5d^1 6s^2$
- (3) $[\text{Xe}] 4f^4 6s^2$
- (4) $[\text{Xe}] 4f^5 6s^1$

 16. pOH of mixture containing 0.1 M CH_3COOH and 0.2 M CH_3COONa is [pK_a of $\text{CH}_3\text{COOH} = 4.76$]

- (1) 9.0561
- (2) 5.0610
- (3) 8.9390
- (4) 7.2652

17. The correct order of limiting molar conductivity for given ions in water at 298 K is

- (1) $\text{Na}^+ > \text{K}^+ > \text{Mg}^{2+} > \text{Ca}^{2+}$
- (2) $\text{K}^+ > \text{Na}^+ > \text{Ca}^{2+} > \text{Mg}^{2+}$
- (3) $\text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+ > \text{Na}^+$
- (4) $\text{Ca}^{2+} > \text{Mg}^{2+} > \text{Na}^+ > \text{K}^+$

18. Element that does not show variable oxidation state is :

- (1) Chlorine
- (2) Fluorine
- (3) Bromine
- (4) Iodine

19. Choose the correct statement about the hydrides of group 16 elements :

- a. The thermal stability of the hydrides decreases in the order $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
- b. Their acidic character decreases from H_2O to H_2Te
- c. Their reducing character increases from H_2S to H_2Te .

Choose the most appropriate answer from the option given below :

- (1) a and b only
- (2) b and c only
- (3) a and c only
- (4) a, b and c

20. Sum of oxidation state of bromine in hypobromous acid and perbromic acid is

- (1) 8
- (2) 10
- (3) 6
- (4) 12

21. Given below are two statements

Statement I : o-nitrophenol is more acidic than m-nitrophenol and p-nitrophenol

Statement II: Propanol will give immediate turbidity with Lucas reagent.

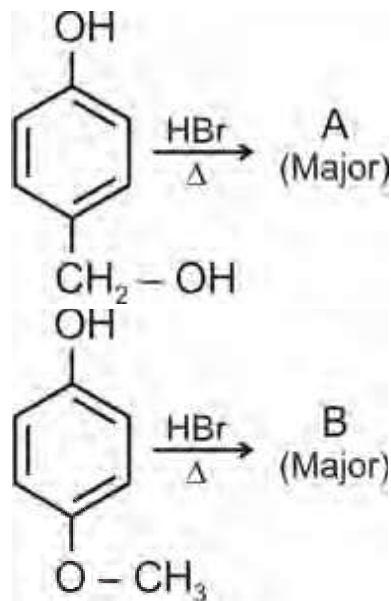
In the light of above statements, choose the correct answer from options given below

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Both statement I and statement II are correct

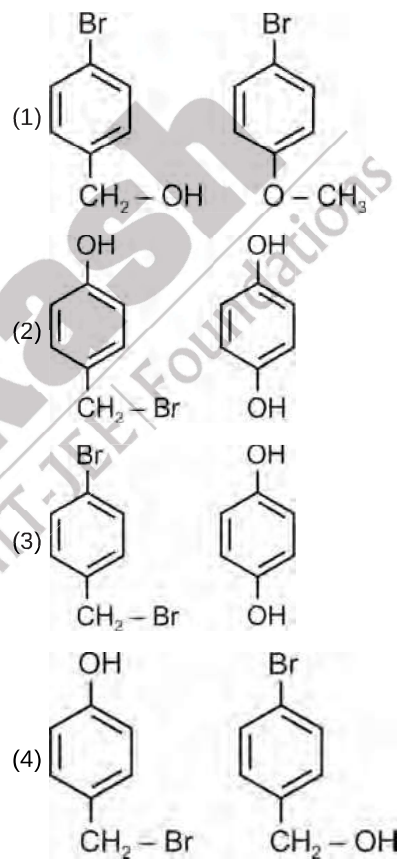
22. Which among the following has highest boiling point?

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- (2) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CHO}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{--OH}$
- (4) $\text{CH}_3\text{--CH}_2\text{--O--CH}_3$

23.



Product 'A' and 'B' formed in the above reactions are



24.

In the hydrocarbon $\overset{1}{\text{CH}_3} - \overset{2}{\text{CH}} = \overset{3}{\text{C}} = \overset{4}{\text{CH}} - \overset{5}{\text{CH}_2} - \overset{6}{\text{CH}_3}$ the state of hybridisation of carbons 2,3 and 5 respectively are

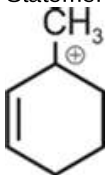
- (1) sp^2 , sp^2 and sp^3
- (2) sp , sp^2 and sp^3
- (3) sp^2 , sp and sp^3
- (4) sp^2 , sp^3 and sp

25. In the kjeldahl method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.56 g of sample neutralises 10 ml of 1 M H_2SO_4 . The percentage of nitrogen in soil is

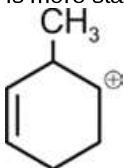
(1) 20%
(2) 28%
(3) 42%
(4) 50%

26. Consider the following statements :

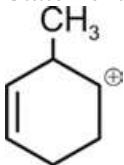
Statement I :



is more stable as compared to



Statement II:



has two hyperconjugation structures

In the light of above statements choose the correct option

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

27. The total number of oxygen atoms involved in proxy linkage in CrO_5 is

(1) 0
(2) 2
(3) 4
(4) 5

28. Reaction of aluminium metal with aqueous sodium hydroxide leads to the formation of

(1) Sodium tetrahydroaluminate (III) and hydrogen gas
(2) Aluminium hydroxide and sodium hydride
(3) Sodium tetrahydroaluminate (II) and hydrogen gas
(4) Aluminium hydroxide and hydrogen gas

29. n-Hexane on heating to 773 K at 10-20 atmospheric pressure in presence of Cr_2O_3 majorly give

(1) Benzene
(2) Cyclohexane
(3) Cyclohexene
(4) 1,4-cyclohexadiene

30. Water soluble and fat soluble vitamins respectively are

(1) Thiamine and Vitamin E
(2) Vitamin C and Riboflavin
(3) Vitamin B_{12} and Vitamin B_6
(4) Vitamin D and Vitamin C

31. Orbital angular momentum of electron present in 3d orbital is

(1) $\sqrt{2} \hbar$
(2) $2\sqrt{3} \hbar$
(3) $\sqrt{6} \hbar$
(4) $\sqrt{3} \hbar$

32. Consider the following compounds

2,2-Dimethylpropane	Butane	2-Methylpropane	2-Methylbutane
(I)	(II)	(III)	(IV)

The correct order of melting point is

(1) (i) > (iii) > (ii) > (iv)
(2) (ii) > (iii) > (iv) > (i)
(3) (iv) > (i) > (ii) > (iii)
(4) (i) > (ii) > (iii) > (iv)

33. Consider the following statements

a. Hormones are molecules that act as intercellular messengers
b. Thyroxine produced in thyroid gland is an iodinated derivative of amino acid proline.
c. Hypothyroidism is characterised by lethargy and obesity.

Choose the correct statements.

(1) a and b only
(2) b and c only
(3) a and c only
(4) a, b and c

34. Total energy of electron in the first excited state of Li^{2+} ion is

(1) $-5.4 \times 10^{-19} \text{ J}$
(2) $-8.7 \times 10^{-18} \text{ J}$
(3) $-9.8 \times 10^{-18} \text{ J}$
(4) $-4.9 \times 10^{-18} \text{ J}$

35. Highest occupied molecular orbital (HOMO) of N_2 molecule is

- (1) π^*2p_x
- (2) σ^*2p_z
- (3) $\sigma 2p_z$
- (4) $\sigma 2s$

36. In which of the following molecule $p\pi-d\pi$ bond is present?

- (1) N_2
- (2) SO_3
- (3) CO_2
- (4) C_2H_2

37. Consider the following statements about PCl_5 molecule

- a. P has sp^3d hybridisation
 - b. It has trigonal bipyramidal structure
 - c. All P-Cl bonds are equal in length
- The correct statements are

- (1) a and b only
- (2) b and c only
- (3) a and c only
- (4) a, b and c

38. An ideal solution has 1 : 3 molar ratio of hexane and benzene. If vapour pressure of hexane and benzene at 298 K are 360 mm and 300 mm Hg respectively, then the mole fraction of hexane in vapour phase would be

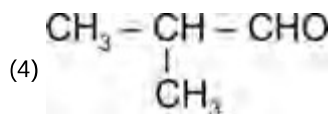
- (1) 0.35
- (2) 0.28
- (3) 0.55
- (4) 0.72

39. Under the constant pressure and temperature condition, most soluble gas in water will be (K_H : Henry's constant)

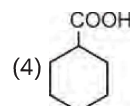
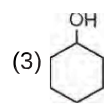
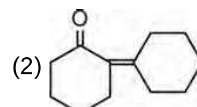
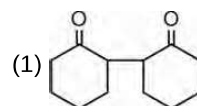
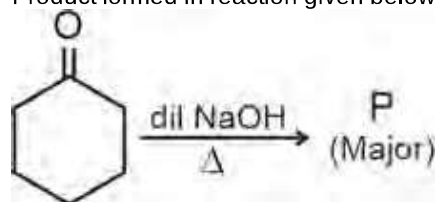
- (1) He ($K_H = 144.17$ kbar)
- (2) H_2 ($K_H = 69.16$ kbar)
- (3) N_2 ($K_H = 76.48$ kbar)
- (4) CH_4 ($K_H = 0.413$ kbar)

40. Which among the following is most reactive towards HCN?

- (1) CH_3CHO
- (2) $HCHO$
- (3) C_6H_5CHO



41. Product formed in reaction given below is,



42. Color of solution formed when dimethyl glyoxime is added to the aqueous solution of nickel chloride and is made alkaline by adding NH_4OH solution, will be

- (1) Black
- (2) Red
- (3) Yellow
- (4) Blue

43. Time of completion of zero order reaction with initial concentration 'a M' and rate constant k is

- (1) $\frac{a}{k}$
- (2) $\frac{2a}{k}$
- (3) $\frac{a}{2k}$
- (4) ak

44. Match compounds List-I with their colour in List-II

List-I	List-II
a. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$	(i) Green
b. PbS	(ii) Prussian blue
c. $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	(iii) Blood red
d. $\text{Fe}(\text{SCN})_3$	(iv) Black

Correct match is

- (1) a(iv), b(ii), c(i), d(iii)
 (2) a(ii), b(iv), c(i), d(iii)
 (3) a(iv), b(ii), c(iii), d(i)
 (4) a(ii), b(iv), c(iii), d(i)

45. Unit of rate of reaction for second order reaction is

- (1) s^{-1}
 (2) $\text{mol}^{-1} \text{L s}^{-1}$
 (3) $\text{mol L}^{-1} \text{s}^{-1}$
 (4) $\text{mol L}^{-1} \text{s}$

PHYSICS

46. Consider the following statements and tick the correct option.

Statement A: For an object moving in uniform circular motion its speed and acceleration, both remain constant.

Statement B: Acceleration and speed, both are vector quantities.

- (1) Statement A is correct while statement B is incorrect.
 (2) Statement B is correct while statement A is incorrect.
 (3) Both statement A and statement B are correct.
 (4) Both statement A and statement B are incorrect.

47. In which process, P - V diagram is a straight line parallel to pressure axis?

- (1) Isochoric
 (2) Isobaric
 (3) Isothermal
 (4) Adiabatic

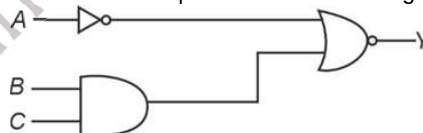
48. In a sample of hydrogen gas, atoms are excited to the stationary state designated by the principal quantum number $n = 5$. The number of possible spectral lines emitted by these atoms, according to Bohr's theory will be

- (1) 6
 (2) 10
 (3) 8
 (4) 5

49. Which of the following is acceptor impurity element?

- (1) Aluminium
 (2) Germanium
 (3) Gallium
 (4) Both (1) and (3)

50. The Boolean equation for the circuit given below is



- (1) $\bar{A} \cdot (B + C)$
 (2) $A \cdot (\bar{B} + C)$
 (3) $A \cdot (\bar{B} + \bar{C})$
 (4) $A \cdot (B + C)$

51. When a beta-positive particle (e^+) is emitted from a nucleus, its neutron-proton ratio will

- (1) Decrease
 (2) Increase
 (3) Remain same
 (4) Either increase or decrease

52. Ammeter and voltmeter readings were recorded as 2 A and 1 V during the experiment to determine the resistance of a given wire using ohm's law. The correct value of the resistance is

(1) $1\ \Omega$
 (2) $2\ \Omega$
 (3) $0.5\ \Omega$
 (4) $5\ \Omega$

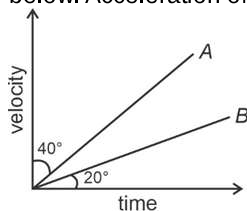
53. The dimensional formula of heat supplied to the system is

(1) $[ML^2T^{-2}]$
 (2) $[ML^1T^{-2}]$
 (3) $[MLT^{-1}]$
 (4) $[ML^2T^{-1}]$

54. The displacement of a particle moving along y-axis with respect to time is $y = at + bt^2 - ct^3$. The dimension of b is same as dimension of

(1) Velocity
 (2) Acceleration
 (3) Displacement
 (4) Force

55. The velocity-time graph for two bodies A and B are shown below. Acceleration of body A and B are in the ratio?



(1) $\frac{\tan 50^\circ}{\tan 20^\circ}$
 (2) $\frac{\tan 20^\circ}{\tan 30^\circ}$
 (3) $\frac{\tan 40^\circ}{\tan 20^\circ}$
 (4) $\frac{\tan 20^\circ}{\tan 40^\circ}$

56. A body is travelling east with a speed of 9 m/s and with an acceleration of 2 m/s^2 acting west on it. The displacement of the body during the 4th second of its motion is

(1) 1 m
 (2) 2 m
 (3) 6 m
 (4) Zero

57. A stationary body of mass 3 kg explodes into three equal pieces. Two of the pieces fly off in two mutually perpendicular directions, one with a velocity of $3\hat{i}$ m/s and the other with a velocity of $4\hat{j}$ m/s. If the explosion occurs in 10^{-4} s, the average force acting on the third piece in newton is

(1) $-(3\hat{i} + 4\hat{j}) \times 10^{-4}\text{ N}$
 (2) $-(3\hat{i} + 4\hat{j}) \times 10^4\text{ N}$
 (3) $(4\hat{i} - 3\hat{j}) \times 10^{-4}\text{ N}$
 (4) $-(4\hat{i} + 3\hat{j}) \times 10^4\text{ N}$

58. A car moves at an optimum speed of 20 m/s on a banked track while describing an arc of a circle of radius $40\sqrt{3}$ m. The angle of banking is ($g = 10\text{ m/s}^2$)

(1) 15°
 (2) 30°
 (3) 60°
 (4) 75°

59. Consider two vectors $\vec{A} = 2\hat{i} + 2\hat{j}$ and $\vec{B} = \hat{i} + c\hat{j}$. If angle between $\vec{A}$ and $\vec{B}$ is 90° , then the value of c is

(1) -1
 (2) 1
 (3) Zero
 (4) 2

60. Consider the given statements and choose the correct option.

Statement A: The kinetic energy of a body depends on the frame of reference.

Statement B: As mass m and v^2 or $\vec{v} \cdot \vec{v}$ in the expression of kinetic energy are always positive, so K.E. can never be negative.

Statement C: Work done by a force along a closed path is always non-zero.

(1) Only statements A and B are true
 (2) Only statements B and C are true
 (3) Only statements A and C are true
 (4) All statements are true

61. Mutual inductance has same unit as

(1) Self-inductance
 (2) Capacitance
 (3) Resistance
 (4) Impedance

62. A particle is executing SHM on a line 4 cm long. If its velocity at mean position is 12 cm/s, its frequency in hertz will be

- (1) $\frac{2\pi}{3}$
- (2) $\frac{3}{2\pi}$
- (3) $\frac{\pi}{3}$
- (4) $\frac{3}{\pi}$

63. A closed organ pipe of length L vibrating in first overtone, is shown in figure given below.



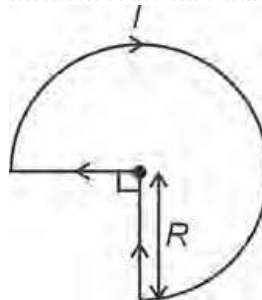
$X = 0$ $X = L$

Match the physical quantities given in column I with their position (or value) given in column II.

	Column I		Column II
a.	Displacement node	(i)	$\frac{L}{3}$
b.	Displacement antinode	(ii)	$\frac{4L}{3}$
c.	Wavelength	(iii)	$\frac{2L}{3}$

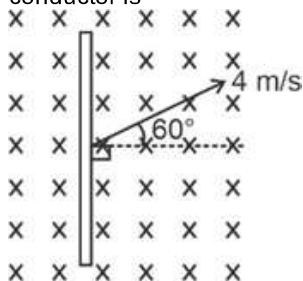
- (1) a(iii), b(ii), c(i)
 - (2) a(i), b(ii), c(iii)
 - (3) a(i), b(iii), c(ii)
 - (4) a(iii), b(i), c(ii)
64. Which of the following statement is incorrect about electromagnetic waves?
- (1) They are transverse in nature
 - (2) They can be produced by accelerating charges
 - (3) They travel with the same speed in all media
 - (4) They travel in free space with the speed of light 3×10^8 m/s
65. If the amplitude ratio of two sources producing interference is 3 : 5, then the ratio of intensities at maxima and minima is
- (1) 25 : 16
 - (2) 5 : 3
 - (3) 16 : 1
 - (4) 25 : 9

66. A current carrying wire in the form of a circular arc of radius R carrying current I is shown in figure. The magnetic moment of the wire will be



- (1) $\pi R^2 I$
 - (2) $\frac{\pi R^2 I}{4}$
 - (3) $\frac{3\pi R^2 I}{4}$
 - (4) $\frac{\pi R^2 I}{2}$
67. Consider the following statements A and B and identify the correct answer:
Statement A: To convert a galvanometer into a voltmeter, one should connect a high resistance in series with galvanometer.
Statement B: To convert a galvanometer into an ammeter, one should connect a low resistance in parallel with galvanometer.
- (1) Both statements A and B are correct
 - (2) Both statements A and B are incorrect
 - (3) Statement A is correct but statement B is incorrect
 - (4) Statement A is incorrect but statement B is correct
68. A magnetic needle suspended parallel to a magnetic field required 4 J of work to turn it through 53° . The torque needed to maintain the needle in this position is
- (1) 4 N m
 - (2) 5 N m
 - (3) 10 N m
 - (4) 8 N m
69. A 40 mH inductor is connected to a $V = 40 \sin(500t)$ volt A.C. supply. The peak value of current in the inductor will be
- (1) 1 A
 - (2) 2 A
 - (3) 3 A
 - (4) 4 A

70. A straight conductor of length 0.6 m is moved with speed 4 m/s perpendicular to a uniform magnetic field of intensity 2 Wb/m^2 as shown in figure. The induced e.m.f. across the conductor is



- (1) 1.2 V
(2) 3.6 V
(3) 2.4 V
(4) Zero
71. A step-down transformer connected to a D.C. mains 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in secondary circuit?

- (1) 4 A
(2) 1 A
(3) 2 A
(4) Zero

72. Given below are two statements:

Statement I: For an elastomer, stress vs. strain graph is a straight line.

Statement II: Steel is an example of elastomer.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct.
(3) Both statement I and statement II are correct.
(4) Both statement I and statement II are incorrect.

73. Two particles of masses $3m$ and $5m$ are at a distance d apart. Due to their mutual gravitational force, they start moving towards each other. Choose the correct statement when they are $\frac{d}{4}$ distance apart.

- (1) Acceleration of mass $3m$ is $\frac{48 Gm^2}{d^2}$
(2) Acceleration of mass $5m$ is $\frac{80 Gm^2}{d^2}$
(3) Acceleration of their centre of mass is zero
(4) All of these

74. A particle is moving parallel to x-axis with constant velocity, then about origin:

- (1) Torque acting on particle is constant and non-zero
(2) Angular momentum of particle is zero
(3) Angular momentum of particle is constant
(4) Angular acceleration of particle is variable

75. There is a system of six particles each of mass m placed at the vertices of a regular hexagon having side l . Potential at the centre of hexagon is

- (1) Zero
(2) $\frac{-3 Gm}{l}$
(3) $\frac{-6 Gm}{l}$
(4) $\frac{-2\sqrt{3} Gm}{l}$

76. In the given figure, earth goes around the sun in elliptical orbit. The product $(|\vec{r} \times \vec{v}|)$ of velocity of earth ($\vec{v}$) and

its position from sun ($\vec{r}$) will be



- (1) Maximum at A
(2) Maximum at B
(3) Maximum at C
(4) Same at all points

77. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Reflecting telescopes use a concave mirror rather than a lens for the objective.

Reason (R): There is no chromatic aberration in a mirror.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

78. An observer on ground, near the surface of lake, sees a bird flying in the air and a fish swimming in water ($\mu = \frac{4}{3}$). At an instant, the fish and the observer see the bird at a height h_1 and h_2 from water surface respectively. Then value of $\frac{h_1}{h_2}$ is

- (1) $\frac{25}{12}$
 (2) $\frac{3}{4}$
 (3) $\frac{8}{3}$
 (4) $\frac{4}{3}$

79. Consider the following statements:

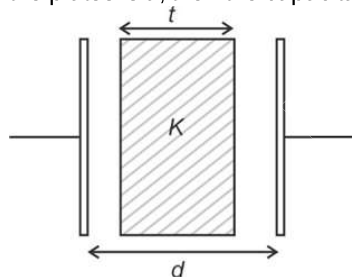
Statement (1) : Resistivity of any conductor is the property of material, while resistance of any conductor doesn't depend on the property of the material.

Statement (2) : Resistivity of any conductor varies with temperature.

Mark the correct option.

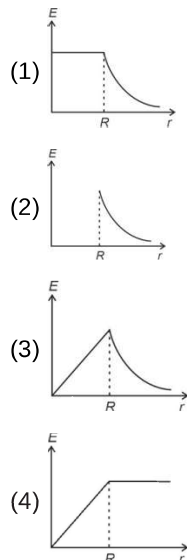
- (1) Only statement (1) is correct.
 (2) Only statement (2) is correct.
 (3) Both statements (1) and (2) are correct.
 (4) Both statements (1) and (2) are incorrect.

80. A dielectric slab of dielectric constant K and thickness t is placed between the plates of a parallel plate capacitor. If area of each plate of capacitor is A and distance between the plates is d , then the capacitance of the capacitor is



- (1) $\frac{A\epsilon_0 K}{Kd+t(1+K)}$
 (2) $\frac{A\epsilon_0 K}{d+t(1-K)}$
 (3) $\frac{A\epsilon_0}{Kd+t(1-K)}$
 (4) $\frac{A\epsilon_0 K}{Kd+t(1-K)}$

81. A non-conducting solid sphere of radius R is uniformly charged. Plot of magnitude of electric field (E) vs distance from centre (r) curve is best represented by



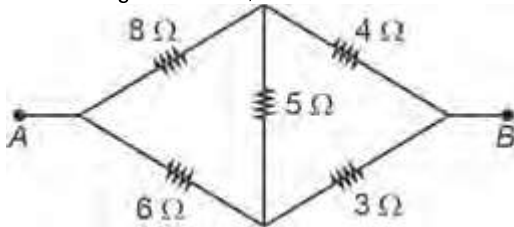
82. Electrical potential V in space as a function of coordinates is given by $V = \frac{1}{x} + \frac{1}{y}$, then the electric field intensity at $(1, 1)$ is given by (in SI units)

- (1) $\hat{i} + \hat{j}$
 (2) $\hat{i} - \hat{j}$
 (3) $2\hat{i} + \hat{j}$
 (4) $2\hat{i} - \hat{j}$

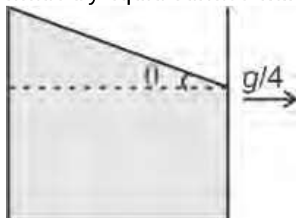
83. Force between two point charges q_1 and q_2 placed at distance d apart is F . The new force between q_1 and q_2 if another charge q_3 is placed between them, will be

- (1) $\left(\frac{q_1+q_2}{q_3}\right)F$
 (2) F
 (3) $\left(\frac{q_3}{q_1+q_2}\right)F$
 (4) $\left(\frac{q_1-q_2}{q_3}\right)F$

84. The equivalent resistance between the points A and B in the circuit given below, is



- (1) $\frac{25}{7} \Omega$
 (2) $\frac{29}{7} \Omega$
 (3) $\frac{36}{7} \Omega$
 (4) $\frac{37}{7} \Omega$
85. Consider the following statements and tick the correct option.
Statement I : Bernoulli's equation can be applied for fluids with zero viscosity.
Statement II : Steady flow is achieved at low flow speeds while beyond a limiting value this flow loses steadiness and becomes turbulent.
- (1) Statement I is correct while statement II is incorrect
 (2) Statement II is correct while statement I is incorrect
 (3) Both statement I and statement II are correct
 (4) Both statement I and statement II are incorrect
86. A container filled with a liquid is accelerating horizontally with acceleration $\frac{g}{4}$ horizontally as shown in figure. Angle θ made by liquid surface will be



- (1) $\tan^{-1} \left(\frac{1}{4} \right)$
 (2) $\tan^{-1} \left(\frac{1}{2} \right)$
 (3) $\tan^{-1}(2)$
 (4) $\tan^{-1}(4)$

87. Two spheres of same material and same surface finish, having radius r and $3r$ are heated to same temperature and are kept in identical surrounding. Ratio of their rate of loss of heat is

- (1) 1 : 2
 (2) 1 : 4
 (3) 1 : 9
 (4) 1 : 3

88. A monoatomic gas at pressure P and volume V is compressed adiabatically to $\frac{1}{27}$ of its original volume. Final pressure of the gas will become

- (1) $64P$
 (2) $27P$
 (3) $9P$
 (4) $243P$

89. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion (A): In photoelectric effect, the number of photoelectrons emitted is always equal to number of photons incident.

Reason (R): Photoelectric effect was explained using wave theory of light.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false

90. On increasing number density of molecules for a gas in a vessel, mean free path of gas

- (1) Decreases
 (2) Increases
 (3) Remains same
 (4) May increase or decrease

BIOLOGY

91. Which of the following is a characteristic of a dicot leaf?

- (1) Same size of vascular bundles
 (2) Reticulate venation
 (3) Absence of stomata on abaxial surface
 (4) Conjoint open vascular bundles

92. Read the given comparative features of two plant specimens 'X' and 'Y' and choose the **correct** option

Character	'X'	'Y'
Hypodermis	Sclerenchymatous	Absent
Endodermis	Absent	Distinct
Xylem	Endarch	Exarch
Bundle sheath cells	Present	Absent

'X' and 'Y' are most likely to be

- (1) X → Gram leaf, Y → Maize stem
 (2) X → Wheat root, Y → Mustard leaf
 (3) X → Grass stem, Y → Mango root
 (4) X → Sunflower root, Y → Maize root
93. Which of the following provide(s) additional stability to the helical structure of DNA over RNA?
- (a) N-glycosidic linkages
 (b) Stacking of one base pair over the other in double helix
 (c) Hydrogen bonding
 (d) Presence of thymine instead of uracil
- Choose the **correct** option

- (1) (a), (b) and (c) only
 (2) (b) only
 (3) (c) and (d) only
 (4) All except (a)

94. Select the **correctly** matched pair w.r.t. the enzymes and their functions.

1.	RNA polymerase III	–	Transcription of snRNA
2.	Nucleases	–	Uncoiling and separation of two strands of DNA
3.	RNA polymerase II	–	Join the two RNA molecules together
4.	Severo Ochoa enzyme	–	Break the DNA into fragments

- (1) (1)
 (2) (2)
 (3) (3)
 (4) (4)

95. Read the given statements of assertion (A) and reason (R) and choose the **correct** option.

Assertion (A) : During DNA replication, both strands of DNA are synthesized in the 3' to 5' direction.

Reason (R) : DNA polymerase can initiate DNA synthesis at 3' end of DNA on its own.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false

96. Which of the following genetic codons codes for the amino acid methionine?

- (1) ACG
 (2) AUC
 (3) AUG
 (4) AUA

97. Observe the following figure



Which of the following depictions regarding the above figure is **correct**?

The given structure represents the living component of (1) phloem which possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus.

The given structure represents a long cylindrical tube-like structure which are interconnected through perforations in their common walls.

- (2) This structure stores food material in the form of starch and fat
 (3) This structure helps only in radial conduction of water

98. State **true (T)** or **false (F)** to the given statements.

(A) One of the hallmarks of the double helical model of DNA proposed by Watson and Crick was base pairing between the two strands of polynucleotide chains.

(B) Avery, MacLeod and McCarty with the help of their experiments concluded that DNA is the hereditary material.

Choose the **correct** option

	(A)	(B)
(1)	T	F
(2)	F	T
(3)	T	T
(4)	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

99. Read the following statements and select the **correct** option

Statement A : Families are characterised on the basis of both vegetative and reproductive features of plant species.

Statement B : Family has a group of related genera with still more number of similarities as compared to genus and species.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

100. Read the following statements (A-D).

(A) *Puccinia* lacks sex organs.

(B) White spots seen on mustard leaves are due to a member of class Phycomycetes.

(C) Mycelium of imperfect fungi is aseptate and coenocytic and large number of them are decomposers of litter and help in mineral cycling.

(D) The asexual spores are generally not found but vegetative reproduction by fragmentation is common in basidiomycetes.

In the light of the above statements, select the option for the **correct** ones.

(1) Only (A), (B) and (D)

(2) Only (C) and (D)

(3) Only (A) and (B)

(4) Only (B), (C) and (D)

101. The prokaryotic organisms that are responsible for the production of biogas from the dung of ruminant animals, are called

(1) Eubacteria

(2) Methanogens

(3) Thermoacidophiles

(4) Halophiles

102. Select the plant that is **odd** one out w.r.t. perigynous flower.

(1) Mustard

(2) Plum

(3) Peach

(4) Rose

103. Match column I with column II and select the **correct** option.

	Column-I		Column-II
a.	Compositae	(i)	Sunhemp
b.	Solanaceae	(ii)	Cotton
c.	Fabaceae	(iii)	Marigold
d.	Malvaceae	(iv)	Chilli

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(ii), b(iii), c(i), d(iv)

104. Which of the following statements is **not** correct?

(1) Aristotle used simple morphological characters to classify plants into herbs, shrubs and trees

The three domains Bacteria, Archaea and Eukarya are

(2) believed to have originated from common ancestor called progenote

Linnaeus classified animals into Anaima and Enaima

(3) respectively on the basis of absence and presence of red blood

According to Whittaker's classification, all prokaryotic

(4) organisms were grouped together under the kingdom Monera

105. Read the following characteristics of family 'X'

(A) Racemose inflorescence

(B) Bisexual and zygomorphic flowers

(C) Gynoecium is monocarpellary, unilocular with many ovules and superior ovary

(D) Sepals are five, with valvate/imbricate aestivation and gamosepalous condition

(E) Many plants belonging to this family are source of pulses.

In the light of above characteristics the family 'X' is

(1) Fabaceae

(2) Solanaceae

(3) Liliaceae

(4) Malvaceae

106. In humans, the visceral nervous system consists of

- (a) Nerves
- (b) Nerve fibres
- (c) Ganglia
- (d) Plexuses

Select the correct option.

- (1) Only (a) and (c)
- (2) Only (a), (b) and (c)
- (3) Only (c) and (d)
- (4) (a), (b), (c) and (d)

107. There are 4 DNA molecules labelled as A, B, C and D respectively. Amongst them, the molecular weight of B is twice of that of D, whereas A is the smallest in size. Size of C is bigger than A, but is still smaller than D.

If we perform gel electrophoresis with these 4 molecules of DNAs, the correct order in which they will separate on the agarose gel from anode to cathode is

- (1) A, B, C, D
- (2) D, C, B, A
- (3) C, A, D, B
- (4) A, C, D, B

108. While performing an experiment, exonucleases were added to a tube containing plasmids and chromosomal DNAs of an eukaryotic cell. What can be the possible implication of this?

- (1) Exonucleases will cut the plasmids into many fragments
- (2) Since the chromosomal DNA is always circular, exonucleases will not act on it
- (3) Both plasmids and chromosomal DNA will remain unaffected by exonucleases
- (4) Exonucleases will cut at the ends of the chromosomal DNA

109. Replication of pBR322 will be affected when an alien gene will be added at the site of restriction enzyme

- (1) *Cla* I
- (2) *EcoR* I
- (3) *Pst* I
- (4) *Pvu* II

110. Match column I with column II w.r.t. human skeletal system.

Column I	Column II
a. Cartilaginous joint	(i) Knee joint
b. Pivot joint	(ii) Sutures present in cranium
c. Fibrous joint	(iii) Between atlas and axis
d. Hinge joint	(iv) Pubic symphysis

Choose the correct option

- (1) a(iv), b(iii), c(i), d(ii)
- (2) a(iv), b(ii), c(i), d(iii)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(i), b(ii), c(iii), d(iv)

111. Read the following statements carefully

- (a) Interstitial fluid has the same mineral distribution as that of plasma.
- (b) Tissue fluid helps in exchange of gases.
- (c) Fats are absorbed through blood in lacteals.

Choose the correct option.

- (1) Only statement (c) is correct
- (2) Only statements (a) and (b) are correct
- (3) All statements (a), (b) and (c) are correct
- (4) Only statements (b) and (c) are correct

112. In human skeletal system, a cavity which is found at the point of fusion of ilium, ischium and pubis is called

- (1) Glenoid cavity
- (2) Acetabulum
- (3) Buccal cavity
- (4) Sella tursica

113. Choose the correct set of flat bones in human skeletal system

- (1) Ribs, scapula, humerus
- (2) Metacarpal, femur, ribs
- (3) Femur, humerus, scapula
- (4) Ribs, scapula, sternum

114. All of the following are the components of serum, **except**

- (1) Clotting factors
- (2) Glucose
- (3) Antibody
- (4) Minerals like Na^+ , Ca^{++}

115. The correct set of hormones that regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome is

- (1) Glucagon, thyroxine, adrenaline
- (2) Prolactin, cortisol, insulin
- (3) Estradiol, triiodothyronine, cortisol
- (4) Oxytocin, epinephrine, somatostatin

116. Select the **correct** statement regarding the group of plants in which ovules are not enclosed within the ovary wall

- (1) They produce microspores and megaspores, within a structure that is haploid

The pollen grain germinates and pollen tube releases the male gamete into the ovule where it fuses with the egg cell in archegonia

- (2) The functional megaspore enclosed within the nucellus undergoes repeated reductional division and develops into a multicellular female gametophyte

Like seedless vascular plants, members of this group have independent free living male and female gametophyte

- 117.** The members of Rhodophyceae are commonly called red algae because of the predominance of _____ in their body.

Select the **correct** option to fill in the above blank.

- (1) Carotene
- (2) r-phyococyanin
- (3) Fucoxanthin
- (4) r-phycoerythrin

- 118.** Identify the option that **incorrectly** depicts the product and its application, associated with a particular microbe.

	Microbe	Product	Application
(1)	<i>Streptococcus</i>	Streptokinase	Used as clot bluster
(2)	<i>Acetobacter aceti</i>	Citric acid	Preparation of Vinegar
(3)	<i>Trichoderma polysporum</i>	Cyclosporin A	Used as immunosuppressive agent in organ transplant patients
(4)	<i>Monascus purpureus</i>	Statins	Blood-cholesterol lowering agents

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

- 119.** If '+' sign is assigned to beneficial interaction, '-' sign to detrimental and '0' sign to neutral interaction then the correct representation for competition, amensalism and parasitism respectively is

	Species A, Species B	Species A, Species B	Species A, Species B
(1)	+, -	-, +	+, 0
(2)	+, +	-, 0	+, -
(3)	-, -	+, 0	+, +
(4)	-, -	-, 0	-, +

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

- 120.** The non-vascular embryophytes

- (a) Have photosynthetic free-living gametophyte
- (b) Show great ecological importance as some of its members are responsible for decomposition of rocks to make the substrate suitable for the growth of higher plants.
- (c) Have main plant body differentiated into foot, seta and capsule.
- (d) Show vegetative reproduction through fragmentation gemmae, and budding

The **correct** ones are

- (1) Only (a) and (c)
- (2) Only (b) and (d)
- (3) Only (b), (c) and (d)
- (4) Only (a), (b) and (d)

- 121.** Which microbe forms symbiotic association with plants of family Fabaceae by forming root nodules?

- (1) *Azospirillum*
- (2) *Acetobacter*
- (3) *Rhizobium*
- (4) *Glomus*

- 122.** The number of births during a given period in the population that are added to the initial density is termed as

- (1) Emigration
- (2) Mortality
- (3) Natality
- (4) Immigration

- 123.** The relationship between species richness and area turns out to be a rectangular hyperbola for a wide variety of organisms and this relationship is a straight line on a logarithmic scale.

Which of the following statements is **true** regarding the relationship mentioned above?

- (1) Species richness keeps on increasing infinitely with increasing explored area within a region
- (2) The regression coefficient is found to be 1.15 for frugivorous birds and mammals in the tropical forests of different continents
- (3) Within a region, slope of the line lies in the range of 1.2 to 1.6 for entire continents
- (4) The slope will become much steeper for very small areas

- 124.** Identify the **odd** one w.r.t. broadly utilitarian ecosystem services of biodiversity.

- (1) Oxygen
- (2) Pollination
- (3) Erosion control
- (4) Firewood

125. Mendel proposed that something stably passed from parent to offspring through gametes and he called these things as 'factors'. These factors regulating the characters are found in pairs and we called them

- (1) Alleles
- (2) Phenotypes
- (3) Pleiotropy
- (4) Chromosomes

126. In genetics, linkage can be described as

- (1) The process where alleles on different chromosomes are inherited together
- (2) The physical association of genes on a chromosome
- (3) Interaction between multiple genes to determine a trait
- (4) Separation of alleles during meiosis

127. Read the statements given below and choose the **correct** option.

Statement I : In case of incomplete dominance, F_1 progenies did not resemble either of the parents and is in between the two.

Statement II : In garden pea, a monohybrid cross considering size of starch grain as character will yield same phenotypic and genotypic ratio in F_2 generation.

- (1) Both the statements I and II are true
- (2) Both the statements I and II are false
- (3) Statement I is true but statement II is false
- (4) Statement I is false but statement II is true

128. Which of the following terms does not belong to the basal bulged part of the pistil?

- (1) Locule
- (2) Septa
- (3) Endothecium
- (4) Placenta

129. In flowering plants, which structure is referred to as the synergid?

- (1) An accessory cell of the pollen grain
- (2) A cell that assists the pollen tube entry into the ovule
- (3) A cell containing two polar nuclei
- (4) A cell that aids in microsporogenesis

130. Read the following Assertion (A) and Reason (R) statements and mark the **correct** option.

Assertion (A): Xenogamy brings genetic variability in the plants.

Reason (R): Transfer of pollen grains from the anther to the stigma of another flower of the same plant takes place during xenogamy.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

131. In a cancer patient, biological response modifiers function by

Activating the immune system and help in destroying

- (1) tumors that avoid detection and destruction by immune system.
- (2) Triggering the activity of cancerous cells
- (3) Lethally irradiating tumor cells
- (4) Decreasing the number of helper T-lymphocytes

132. Saliva in mouth belongs to category of barrier of innate immunity that prevents the entry of microorganisms in human body. Another example that belongs to same category is

- (1) Mucus coating in lining GIT
- (2) Acid in stomach
- (3) Macrophages in tissues
- (4) Interferons

133. Cerebellum is concerned with

- (a) Integration of informations received from auditory system
 - (b) Control of cardiovascular reflexes
 - (c) Regulation of sexual behaviour
- Choose the correct option

- (1) (a) only
- (2) (a) and (b) only
- (3) (b) and (c) only
- (4) (a), (b) and (c)

134. The hormone secreted by α and β cells of Islets of Langerhans of pancreas can be differentiated from each other as

- (1) Former is a peptide hormone while latter is a steroid hormone.
- (2) Former stimulates glycogenolysis while latter stimulates glycogenesis.
- (3) Former acts on hepatocytes and adipocytes while latter acts on hepatocytes only.
- (4) Former enhances cellular glucose utilisation while latter reduces it.

135. A type of epithelium whose function is to move particles or mucus in a specific direction over the epithelium is present in

- (1) Inner surface of fallopian tube
- (2) Lining of stomach
- (3) Proximal convoluted tubule
- (4) Wall of blood vessels

136. Post-industrialisation the increased number of melanised moths in England represents which type of natural selection?

- (1) Stabilising
- (2) Directional
- (3) Disruptive
- (4) Artificial

137. In which district of Madhya Pradesh, pre-historic cave paintings can be seen at Bhimbetka rock shelter?

- (1) Gwalior
- (2) Bhopal
- (3) Vidisha
- (4) Raisen

138. The filtration slits are formed by the cellular arrangement of which of the following cells that are also involved in ultrafiltration?

- (1) Endothelium of glomerular blood vessels
- (2) Basement membrane of renal tubule
- (3) Podocytes of Bowman's capsule
- (4) Endothelium of afferent arteriole

139. Which of the following vertebrates possesses two pairs of limbs, tympanum and only one ventricle in its heart?

- (1) *Bufo*
- (2) *Ichthyophis*
- (3) *Naja*
- (4) *Vipera*

140. Refer the following statements.

- (A) When we see objects in our immediate surroundings we apparently are peeping into the past.
- (B) Huge clusters of galaxies comprise the universe which is almost 20 million years old.
- (C) The Big Bang theory talks of a huge explosion unimaginable in physical terms.
- (D) Earth was supposed to have been formed about 4.5 bya.

Which of the above mentioned statements are not true?

- (1) (A) and (C)
- (2) (B) and (D)
- (3) (C) and (D)
- (4) (A) and (B)

141. Complete the analogy by selecting the correct option.

Thorns of *Bougainvillea* and tendrils of *Cucurbita* : Homology : : Wings of butterfly and that of birds : ____.

- (1) Homology
- (2) Adaptive radiation
- (3) Divergent evolution
- (4) Analogy

142. Match column I with column II

	Column I		Column II
a.	Glycosuria	(i)	Inflammation of glomeruli
b.	Gout	(ii)	Mass of crystallised salts within the kidney.
c.	Renal calculi	(iii)	Presence of glucose in urine
d.	Glomerulo-nephritis	(iv)	Accumulation of uric acid crystals in joints

Choose the correct option

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(ii), b(iii), c(i), d(iv)
- (3) a(iii), b(iv), c(ii), d(i)
- (4) a(iii), b(ii), c(iv), d(i)

143. A child experiencing recurrent infections since birth due to impaired immune system, was taken to the hospital, where after many investigations, he/she was diagnosed with lack of gene for ADA. All of the following are possible treatment options in this scenario, **except**

- (1) Gene therapy with functional ADA gene
- (2) α -1-antitrypsin injections
- (3) Enzyme replacement therapy
- (4) Bone marrow transplantation

144. Assertion (A): For a biotech company to launch a new transgenic product in Indian market, they need approval from GEAC.

Reason (R): Genetic Engineering Appraisal Council is set up by Indian government to make decisions regarding safety of introducing GM-organisms for public services. In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true, but (R) is false

145. *Aurelia* belongs to the phylum, whose members possess

- (1) Choanocytes
- (2) Cnidocytes
- (3) Comb plates
- (4) Collar cells

146.How many documented varieties of Basmati rice, distinctly known for its unique aroma and flavour are grown in India?

- (1) 2,00,000
- (2) 20,000
- (3) 27
- (4) 12

147.Given below are few statements.

- (a) In *Aplysia*, organ system level of organisation is seen.
 - (b) All organisms with tube within tube body plan have triploblastic condition.
 - (c) *Meandrina* has a blind sac body plan.
- Select the correct option

- (1) (a), (b) and (c) are correct
- (2) Only (a) and (b) are correct
- (3) Only (b) and (c) are correct
- (4) Only (a) is correct

148.All of the following cells are present in the main circulating fluid of human body, **except**

- (1) Thrombocytes
- (2) Osteocytes
- (3) Erythrocytes
- (4) Leucocytes

149.The activity of an enzyme can not be affected by a change in which of the factors given below?

- (1) Temperature
- (2) pH
- (3) Substrate concentration
- (4) Activation energy

150.Choose the odd one w.r.t. nucleosides present in the genetic material of causative agent of AIDS.

- (1) Adenosine
- (2) Uridine
- (3) Thymidine
- (4) Cytidine

151.Consider the given diagram and select the correct option w.r.t. it.



- (1) Represents tertiary structure of the protein
- (2) Possesses covalent bonds, disulphide bonds and glycosidic bonds
- (3) It represents beta-plated structure of a protein
- (4) Represents alpha helix structure of a protein

152.Select the incorrect statement.

- The rate of enzyme catalysed reaction doubles or
- (1) decreases by half for every 10°C change in temperature in either direction.
- (2) Rate of a physical or chemical process refers to the amount of product formed per unit time.
- (3) Thermal stability is an important quality of enzymes isolated from thermophilic organisms.
- (4) A chemical change simply refers to a change in shape without breaking of bonds.

153.The type of muscle fibres present in biceps are similar to the type of muscle fibres present in heart in which of the following features?

- (1) Branched appearance
- (2) Uninucleated condition
- (3) Presence of striations
- (4) Absence of communication junctions

154.Ovarian follicles in different stages of development are found embedded in

- (1) Oviduct
- (2) Ovarian ligament
- (3) Ovarian stroma
- (4) Ovarian epithelium

155.The process of childbirth is called parturition which is induced by a complex neuroendocrine mechanism involving all of the following hormones, **except**

- (1) Cortisol
- (2) Prolactin
- (3) Estrogen
- (4) Oxytocin

156.Which plant hormone is known for its role in seed germination as well as stem elongation?

- (1) Zeatin
- (2) Gibberellin
- (3) ABA
- (4) Cytokinin

157.Read the statements given below and select the **correct** option.

Statement I : The primary role of ethylene is to inhibit leaf abscission in a plant.

Statement II : ABA stimulates the closure of stomata to prevent the loss of water in stress conditions.

- (1) Both the statements are true
- (2) Both the statements are false
- (3) Statement I is true but statement II is false
- (4) Statement I is false but statement II are true

158.Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.

Assertion (A): Mendel's law of independent assortment states that when two pairs of traits are combined in a hybrid, segregation of one pair of traits is independent of the other pair of traits.

Reason (R): The principle of independent assortment applies only to the genes that are located closely on the same chromosome.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

159.Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Snake	(i)	Primary carnivore
b.	Cow	(ii)	Converters or transducer
c.	Frog	(iii)	Third order consumer
d.	Phytoplankton	(iv)	Key industry animal

- (1) a(iv), b(i), c(iii), d(ii)
- (2) a(iii), b(i), c(iv), d(ii)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(iv), b(iii), c(ii), d(i)

160.First irreversible reaction of glycolysis where a molecule of ATP is utilised, is catalysed by

- (1) Phosphofructokinase
- (2) Aldolase
- (3) Enolase
- (4) Hexokinase

161.For the synthesis of one sucrose molecule in C_4 plants in comparison to C_3 plants, how many additional ATP molecules are required?

- (1) 60
- (2) 24
- (3) 30
- (4) 12

162.Consider the following Assertion (A) and Reason (R) and choose the correct option

Assertion (A): In plants, like maize, photorespiration does not occur.

Reason (R): In bundle sheath cells of maize leaves, C_4 acid is broken down and releases CO_2 , which ensures that the RuBisCO functions as carboxylase

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

163.Trophic levels in an ecological pyramid represent

- (1) Only the consumers in food chain
- (2) Each level in food chain where transfer of energy takes place
- (3) Vertical distribution of different species in an ecosystem occupying different levels
- (4) The network of interconnected food chains

164.When fatty acid such as tripalmitin is used as a substrate in respiration, the value of RQ is found to be less than one because

- (1) The amount of CO_2 evolved is more than the amount of O_2 consumed
- (2) The ratio of the numbers of carbon and hydrogen atoms in this molecule is 1 : 1
- (3) The amount of O_2 consumed is more than the amount of CO_2 evolved
- (4) It contains less number of carbon atoms than oxygen atoms

165. Na^+/K^+ pump in a cell is an example of

- (1) Active transport
- (2) Passive transport
- (3) Osmosis
- (4) Simple diffusion

166. identify the following statements as **True (T)** or **False (F)** and choose the option accordingly.

- (A) Tracheids are found in all categories of vascular plants.
 (B) All enzymes and electron carriers of aerobic respiration are present in the inner membrane of mitochondria.
 (C) Autophagic vacuoles are formed by the union of many primary lysosomes around old and dead organelles.
 (D) Intermediate filaments help in pseudopodia formation.

	A	B	C	D
(1)	T	F	T	T
(2)	T	F	T	F
(3)	F	T	F	T
(4)	F	T	T	F

- (1) (1)
 (2) (2)
 (3) (3)
 (4) (4)

167. Centriole duplication occurs during _____ of cell cycle. Select the **correct** option to fill in the blank.

- (1) G₁ phase
 (2) G₂ phase
 (3) S phase
 (4) G₀ phase

168. Tetrads formation is a characteristic feature of

- (1) Leptotene stage
 (2) Zygotene stage
 (3) Diplotene stage
 (4) Diakinesis stage

169. Match the following columns and choose the **correct** option.

	Column-I		Column-II
a.	Quiescent stage	(i)	Diplotene
b.	Best stage to study morphology of chromosome	(ii)	Pachytene
c.	Dissolution of synaptonemal complex	(iii)	G ₀ phase
d.	Occurrence of crossing over	(iv)	Metaphase

- (1) a(ii), b(iii), c(iv), d(i)
 (2) a(iii), b(iv), c(i), d(ii)
 (3) a(iii), b(iv), c(ii), d(i)
 (4) a(iii), b(ii), c(iv), d(i)

170. Which of the following is **incorrect** w.r.t. nucleolus?

- (1) It is a spherical structure found in the nucleoplasm
 (2) It is not bound by a membrane
 (3) It is the site for ribosomal RNA (rRNA) synthesis
 (4) elaborate nucleoprotein fibres called chromatin is contained

171. Frogs are beneficial for mankind because they

- (1) Live in water as well as on land
 (2) Have eyes containing single unit
 (3) Protect insects and thus the crops
 (4) Eat insects and protect the crops

172. Select the incorrect match w.r.t. humans.

- (1) Colostrum — Milk produced during initial few days of lactation
 (2) Relaxin — Secreted by ovary
 (3) hCG — Antagonistic to LH
 (4) Umbilical cord — Connects foetus to the placenta

173. Read the following two statements. One is mentioned as **Assertion (A)** and other is mentioned as **Reason (R)**.

Assertion (A): During the process of fertilization, only one sperm can enter ovum under normal conditions.

Reason (R): Sperm after binding with its receptor present on zona pellucida layer of the ovum induces changes in the membrane that blocks the entry of additional sperms. In the light of above statements, choose the correct answer from the options given below:

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (2) (A) is true, but (R) is false
 (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (4) (A) is false but (R) is true

174. Choose the incorrect statement w.r.t. an adult female cockroach.

- (1) The abdomen consists of 10 segments.
 (2) 6th, 7th and 8th abdominal segments form a brood pouch.
 (3) Each ovary is formed of a group of eight ovarioles.
 (4) Oviducts of each ovary unite to form vagina.

175. Which of the following is incorrect w.r.t. humans?

- (1) Release of CO₂ is the result of cellular respiration which is an anabolic process.
 (2) Pleural fluid in the pleural cavity reduces the friction on the lung surface.
 (3) Usage of spirometer helps in clinical assessment of pulmonary functions.
 (4) CO₂ is more soluble in blood as compared to O₂.

176. During normal expiration in humans

- (1) Intrapulmonary pressure decreases
- (2) Volume of thoracic cavity decreases
- (3) External inter-costal muscles contract
- (4) Dome-shaped diaphragm becomes flattened

177. India was among the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as social goal. The programmes were initiated in the year

- (1) 1983
- (2) 1953
- (3) 1960
- (4) 1951

178. Which of the following is/are the complications that may occur in later stages of sexually transmitted infections?

- (a) Abortions, still births
- (b) Infertility, ectopic pregnancy
- (c) Pelvic inflammatory diseases
- (d) Cancer of the reproductive tract

Select the option with correct set.

- (1) Only (a) and (b)
- (2) Only (b) and (c)
- (3) (a), (b), (c) and (d)
- (4) (a), (c) and (d) only

179. If the male partner of a couple is impotent but the female is reproductively healthy then which of the following ARTs the couple should opt that involves *in-vitro* fertilisation?

- (1) ICSI
- (2) AI
- (3) IUI
- (4) GIFT

180. How many of the animals mentioned in the box below respire through lungs?

Sycon, Macaca, Aptenodytes, Bufo, Vipera, Fasciola, Gorgonia

Select the correct option.

- (1) Three
- (2) Four
- (3) Five
- (4) Two

